Lattice refinement followed by the rodeo algorithm: a gate-level benchmark of eigenstate-preparation routes on the Hubbard chain

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Resolution refinement transports an eigenstate adiabatically from a coarse to a fine model space along a shift-controlled, gapped path; followed by the rodeo algorithm, a projective energy filter, it is a route to high-fidelity eigenstates that needs neither a classical optimizer nor training measurements. We implement the lattice version of this pipeline, in which every coarse site is resolved into a block of fine sites, for the Hubbard chain refined from four to eight and from eight to sixteen sites and for the $2 \times 2 \to 4 \times 4$ plaquette, with every stage compiled to rotations and CNOTs and validated against exact diagonalization in the conserved $(N_\uparrow, N_\downarrow)$ sector. Three structural results carry beyond the model. The Galerkin coarse Hamiltonian is itself a Hubbard model with $(t, U) \to (t/2, U/2)$ in one dimension, so the hierarchy is self-similar and can start from the filled dimer, a single basis state. The path conserves total spin and the lattice symmetry, so the transport is governed by the lowest excitation in the symmetry sector of the target, Δ_{eff}, which stays of order t while the sector gap collapses to $0.05t$; the refinement time is flat in U and in size, Δ_{eff} is the right filter width, and a coarse state with the wrong quantum numbers cannot be refined at all. And the cost of a fixed-step digital filter scales as $1/\Delta_{\text{eff}}$ per cycle with an accuracy floor set by the step. The benchmark is then run on equal terms: the same Trotterized filter is applied to the prolonged and refined states, to an ordinary adiabatic ramp from the free-fermion determinant, and to single-layer, pruned and adaptively grown (ADAPT-VQE) unitary coupled-cluster states, and the routes are compared by ground-state infidelity against CNOTs per accepted preparation. At sixteen qubits the ramp is the cheapest route at every target, by a factor three over the pruned ansatz and sixteen over the projector path, whose cost is the coarse term conditioned on all antibonding qubits. At thirty-two qubits the premise changes: the coarse eigenstate starts ahead of the determinant, the ramp needs twice the time to keep up, the single-layer ansatz has lost two orders of magnitude, and every filter cycle costs 2×10^5 CNOTs. There the two-local refinement path ties with the ramp at every target from 3×10^{-3} down, the projector path reaches 10^{-3} within a factor two of it instead of sixteen, and the cost of every route is set by the filter cycles it still needs — which is why the refinement path, whose transport delivers the best un-

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filtered input, is the one worth compiling without its conditioned coarse term.

1 Introduction

Preparing an eigenstate of a many-body Hamiltonian on a quantum computer is the step that every subsequent measurement of a correlated system depends on, and the standard routes each face an obstacle at scale: variational preparation pays for a classical optimization loop and is capped by its ansatz [1, 2], adiabatic preparation pays for the inverse gap along its path [3], and projective filters — phase estimation, polynomial filters, the rodeo algorithm [4, 5] — pay $1/p$ for an input of overlap p with the target. *Resolution refinement* [6] was proposed to supply the filter with a good input without an optimizer: an eigenstate of a coarse (low-resolution) Hamiltonian is lifted by a prolongation operator into a finer model space and transported adiabatically along a path that stays gapped because refinement changes the low-energy spectrum only mildly. In Ref. [7] (paper I below) we implemented this chain — prepare a coarse eigenstate, prolong and refine for a time T, filter with the rodeo algorithm — at the gate level for a hierarchy of single-particle bases of the pairing Hamiltonian, and benchmarked it on equal terms against the same filter applied to cheaper inputs. The verdict there was sober: at eight and sixteen qubits an ordinary adiabatic ramp from the Hartree–Fock determinant reached every target accuracy at a fraction of the cost, and an adaptively grown variational ansatz undercut all filtered routes. The case for the refinement stage rested on its freedom from optimization and on coarse eigenstates that are better starts than the determinant — a regime the pairing hierarchy did not exercise.

The present paper carries the same pipeline, the same benchmarks and the same cost accounting to the Hubbard model [8, 9],

$$H = -\sum_{\langle ij\rangle\sigma} t_{ij}\left(c^\dagger_{i\sigma}c_{j\sigma} + \text{h.c.}\right) + U\sum_i n_{i\uparrow}n_{i\downarrow}, \quad (1)$$

for three reasons. The Hubbard model is the canonical strongly correlated fermion problem and the most frequently used benchmark for the quantum simulation of fermions [10, 11, 12, 13], with the one-body hopping channel that the pairing model lacks. It is the setting for which resolution refinement was proposed [6]: a *lattice* refinement in which every coarse site is resolved into a block of fine sites, the prolongation is a product of Givens rotations, and the fine lattice is the natural discretization of a continuum problem. And it is not exactly solvable in general, so the exact-diagonalization (ED) reference that anchored every number of paper I must be produced numerically; we keep that discipline and benchmark every stage against ED in the conserved $(N_\uparrow, N_\downarrow)$ sector, up to the 14 400-dimensional sectors of the sixteen-site chain and the 4×4 plaquette.

Relative to the proposal of Ref. [6], the rodeo literature [4, 5, 14, 15] and the related schemes that build large systems from solved pieces — the hierarchical fusion method [16] and nucleation [17], which grows the domain at fixed resolution — this work contributes three things. (a) A *reference implementation* of lattice refinement: the block prolongation, the Galerkin coarse model in closed form, two refinement paths (a shift-controlled projector path and a two-local un-dimerization path), the Trotterized refinement and the Trotterized controlled evolution of the filter, all compiled to rotations and CNOTs, validated against ED at sixteen and at thirty-two qubits and reproducible from one executable notebook [18]. (b) A *benchmark on equal terms*: the refinement-plus-filter chain is compared with the same filter applied to the unrefined prolonged state, to the free-fermion determinant, to an ordinary adiabatic ramp that switches on U from it, and to the single-layer, the pruned and the adaptively grown (ADAPT-VQE [19]) unitary coupled-cluster states, with the ground-state infidelity as the common observable, matched evolution accuracy, a CNOT budget per accepted preparation, and the minimum cost at a given target — at eight sites (sixteen qubits) and, for the first time for this pipeline, at sixteen sites (thirty-two qubits), where the free-fermion input has begun to degrade. (c) *Three results that transfer*: the self-similarity of the Galerkin hierarchy, which lets the chain of refinements start from a single basis state; the identification of the symmetry-protected gap Δ_{eff} as the quantity that controls both the refinement time and the filter width, together with the diagnostics that expose a coarse state that cannot be refined; and the

scaling of the fixed-step digital filter — cost per cycle $\propto 1/\Delta_{\text{eff}}$, accuracy floor set by the step — which decides, at every size, whether an input that is cheaper but worse can be rescued by cycles.

The outcome of (b) is stated plainly. At sixteen qubits ($U = 4t$) the ordinary adiabatic ramp from the free-fermion determinant is the cheapest route to every target accuracy: 1.9×10^4, 4.0×10^4 and 9.1×10^4 CNOTs for infidelities 10^{-3}, 3×10^{-4} and 10^{-4}, against three times more for the pruned 98-parameter ansatz, four to seven times more for the two-local refinement path and sixteen times more for the projector path, almost entirely in the coarse term conditioned on all antibonding qubits. The reason is specific to the size: the determinant has the same overlap with the target and the same Δ_{eff} as the prolonged coarse eigenstate. At thirty-two qubits this is no longer so: the prolonged coarse ground state starts at $p_0 = 0.87$ against 0.72 for the determinant, the ramp needs $T = 32/t$ instead of $16/t$, the single-layer ansatz has lost two orders of magnitude and an adaptively grown ansatz of two hundred operators barely reaches 10^{-2}. The ramp remains the cheapest route, but the two-local refinement path now ties with it within the sampling error at every target from 3×10^{-3} down (3.2 against 3.0×10^5 CNOTs at 3×10^{-3}, 8.1 against 7.9×10^5 at 10^{-3}), the variational routes cost 1.3–1.9 times more, and the projector path, which delivers the best unfiltered input, costs 1.6–1.9 times more at the two tightest targets instead of sixteen. At this size the cost of every route is dominated by the filter cycles it still needs, each worth 2×10^5 CNOTs, so what decides the comparison is how close to the target the input starts — and that is the quantity the refinement stage controls.

The paper follows the structure of paper I. Section 2 fixes the model, the sector encoding, the ED reference and the symmetry diagnostics; Sec. 3 defines the lattice prolongation, the Galerkin coarse Hamiltonian, the two paths, the filter and the comparators; Sec. 4 reports the sector-exact results, including the symmetry protection and the wrong-sector plaquette; Sec. 5 gives the circuits and the matched comparison at eight sites; Sec. 6 takes the chain $8 \to 16$ to the circuit level; and Sec. 7 discusses what transfers. The Supplemental Material [20] collects the verification material: the adiabatic bound and the response norm of the path, the dimer-seeded hierarchy and the composition of steps, the plaquette spectra, the variational tables, the Trotterized refinement at eight sites, the exact-ensemble filter, the ingredients of the thirty-two-qubit study and the operating-point tables.

2 Model, reference and symmetry diagnostics

We use $t_{ij} = t = 1$ on nearest-neighbor bonds, open boundary conditions throughout, the chain of L sites and the $L \times L$ square lattice with site index $i = xL + y$. The Hamiltonian conserves $N_\uparrow$ and $N_\downarrow$ separately and all calculations are done in one sector of dimension $\binom{N}{N_\uparrow}\binom{N}{N_\downarrow}$.

Encoding and ED. Spin-orbitals are Jordan–Wigner ordered in two blocks, the N up orbitals followed by the N down orbitals. A down hop then picks up the parity of all up electrons twice, which cancels, so the sector Hamiltonian factorizes exactly as

$$H = K_\uparrow \otimes \mathbb{1} + \mathbb{1} \otimes K_\downarrow + U\,\mathrm{diag}(\text{double occupancy}), \tag{2}$$

with K_σ the one-species hopping matrix on the sorted occupation masks of that species. Reference energies and eigenvectors come from dense diagonalization up to a few thousand states and from ARPACK Lanczos beyond; the Kronecker structure is also what makes every product-formula factor of the thirty-two-qubit study a product of two 120×120 exponentials (Sec. 6). The builder is validated against free fermions at $U = 0$, the analytic dimer $E_0 = \frac{1}{2}(U - \sqrt{U^2 + 16t^2})$, and the term decomposition used for product formulas.

Symmetry diagnostics. Three operators are built in the same sector basis and label every eigenstate. The total spin follows from $\langle \mathbf{S}^2 \rangle = S_z(S_z + 1) + \|S^+\psi\|^2$, with $S^+ = \sum_i c^\dagger_{i\uparrow} c_{i\downarrow}$ applied as a map to the $(N_\uparrow + 1, N_\downarrow - 1)$ sector with the Jordan–Wigner signs of the block ordering. A lattice symmetry is a fermionic site permutation, a signed permutation matrix on each species: the reflection parity for the chain, and the eight elements of C_{4v} for the square, from which the weight of a state in each irreducible representation follows from the character formula

$w_\Gamma = (d_\Gamma/8)\sum_g \chi_\Gamma(g)^* \langle\psi|g|\psi\rangle$. These labels are what decide, in Sec. 4.2, which gap governs the refinement.

3 The pipeline for lattice refinement

3.1 Prolongation

A refinement step maps a coarse lattice of N_c sites onto a fine lattice of $N_f = bN_c$ sites by resolving every coarse site c into a block $B(c)$ of b fine sites, $b = 2$ for the chain (sites $2c, 2c+1$) and $b = 4$ for the square lattice. The prolongation sends the coarse orbital to the normalized symmetric (bonding) combination of its block,

$$c^\dagger_{c\sigma} \longmapsto s^\dagger_{c\sigma} = \frac{1}{\sqrt{b}} \sum_{f\in B(c)} c^\dagger_{f\sigma}, \tag{3}$$

and a coarse Slater determinant to the product of $s^\dagger$ operators; particle numbers are unchanged, so the filling halves (quarters) at each step, as it must for lattices approaching the continuum at fixed particle number [6]. The map P is an isometry from the coarse into the fine sector; its complement is spanned by configurations in which at least one antibonding block mode is occupied. On a register in which the fine qubits of a block are adjacent, P is one Givens rotation per block per spin, two CNOTs each: 16 CNOTs for $4 \to 8$ and 32 for $8 \to 16$. In the sector basis P is built explicitly and $P^\dagger P = \mathbb{1}$ is verified to machine precision.

3.2 The Galerkin coarse Hamiltonian

The coarse Hamiltonian whose eigenstate is refined is the fine Hamiltonian seen at coarse resolution, $H_{\text{low}} = P^\dagger H_{\text{high}} P$. With this choice $\langle P\psi_{\text{low}}|H_{\text{high}}|P\psi_{\text{low}}\rangle = E_{\text{low}}$ holds by construction for the exact coarse eigenstate; it says nothing about the overlap p_0 of the prolonged state with the fine ground state, which must be computed, and it is lost when the coarse state is itself approximate, in which case the coarse error propagates as in paper I — the final infidelity is bounded by $(\sqrt{\epsilon_{\text{low}}} + a_{\text{ad}} + \eta)^2$ with a_{ad} the adiabatic leakage and η the product-formula error — and, as shown in the Supplemental Material [20], the coarse admixture is in practice transported almost unchanged and caps the refined overlap at roughly the coarse fidelity. The Galerkin model has a closed form. Projecting the hopping term with Eq. (3), an intra-block bond contributes a constant $-t$ per particle on the prolonged subspace and an inter-block bond contributes $\frac{1}{2}s^\dagger_c s_{c'}$; projecting the on-site term, $n_{f\sigma} \to n_{s_c\sigma}/b$, gives $(U/b)\, n_{s_c\uparrow} n_{s_c\downarrow}$. Hence

$$\text{chain:}\quad H_{\text{low}} = H_{\text{Hub}}\big(L;\, t/2,\, U/2\big) - tN, \tag{4}$$

$$\text{square:}\quad H_{\text{low}} = H_{\text{Hub}}\big(L\times L;\, t/2,\, U/4\big) - 2tN, \tag{5}$$

with $N = N_\uparrow + N_\downarrow$ and the boundary conditions of the fine lattice. In one dimension U/t is preserved, so a chain of refinements $L \to 2L \to 4L \to \cdots$ is a sequence of Hubbard chains with the same interaction strength and a hopping scale that doubles at each step: the hierarchy is self-similar, and its coarsest member can be the filled dimer, a single computational basis state. In two dimensions the coarse interaction is weaker by a factor of two relative to its hopping. The Galerkin pair $(H_{\text{low}}, H_{\text{high}})$ is a choice; a continuum-scaling pair would start the path from a different coarse ground state and change only p_0.

3.3 Two refinement paths

Path A: the shift-controlled projector path. As in paper I,

$$H_A(s) = \cos^2\!\big(\tfrac{\pi s}{2}\big) P(H_{\text{low}} - \mu)P^\dagger + \sin^2\!\big(\tfrac{\pi s}{2}\big)(H_{\text{high}} - \mu), \qquad s = t/T, \tag{6}$$

with the shift $\mu = E_{\text{low}} + \mu_{\text{buf}}$ placing the complement of the prolonged subspace, where the coarse term vanishes, at energy μ_{buf} above the tracked state at $s = 0$. Because $P(H_{\text{low}} - \mu)P^\dagger = \Pi_a(H_{\text{high}} - \mu)\Pi_a$ with Π_a the projector onto empty antibonding modes, the coarse term is, on a device, the fine Hamiltonian in the bonding/antibonding frame conditioned on all antibonding qubits being in $|0\rangle$ — the multi-controlled structure whose cost dominated the $3 \to 4$ chain of paper I, now with $2N_c$ control qubits.

Path B: adiabatic un-dimerization. The same end points are connected by a path that stays two-local in the laboratory frame: add to the fine Hamiltonian an extra hopping Λ on the

intra-block bonds and switch it off,

$$H_B(s) = H_{\text{high}} - \Lambda \cos^2\!\left(\tfrac{\pi s}{2}\right) \times \sum_{\langle f f' \rangle \in \text{intra}} \sum_{\sigma} \left(c^\dagger_{f\sigma} c_{f'\sigma} + \text{h.c.}\right). \quad (7)$$

At $s = 0$ the antibonding modes are pushed up by 2Λ; for $\Lambda \to \infty$ the ground state of $H_B(0)$ is the prolonged Galerkin ground state, and at finite Λ it differs from it by an admixture of order t/Λ, so a transport that starts from the prolonged state carries a residual $1 - p \simeq c\,(t/\Lambda)^2$ to the end. Path B is the dimerized Hubbard chain being made uniform; every Trotter step along it costs what a step of the fine Hamiltonian costs, with no conditioning.

Refinement time. The exact evolution along either path splits the s interval into M_s slices (96 up to 784 states, 48 with a sparse Krylov exponential for the 14 400-dimensional sectors), diagonalizes or applies the Hamiltonian at the slice midpoint, and reports the overlap $p(T) = |\langle \Psi_{\text{high}} | \psi(T) \rangle|^2$ with the ED ground state on a grid of T; T^* is the first grid value with $1 - p < 10^{-3}$. As in paper I the pipeline is optimizer-free in execution but not parameter-free: T, μ_{buf} (or Λ), the number of product-formula steps n_s, the filter step δt and the objective energy E_{obj} are all chosen with knowledge of the exact answer, and a device-level calibration protocol for them is part of the open work.

3.4 The rodeo filter

The filter is unchanged from paper I: M cycles of the controlled evolution $e^{-iH_{\text{high}}t_k}$ on one reused ancilla with the phase $e^{iE_{\text{obj}}t_k}$, Gaussian cycle times of RMS width σ, and post-selection on the ancilla. For an input $|\psi\rangle = \sum_k c_k |E_k\rangle$ and $E_{\text{obj}} = E_0$ the fidelity of the ensemble of accepted runs obeys exactly

$$1 - \mathcal{F}_M = \frac{\sum_{k \neq 0} |c_k|^2 q_k^M}{p + \sum_{k \neq 0} |c_k|^2 q_k^M}, \qquad q_k = \tfrac{1}{2}\left(1 + e^{-(E_k - E_0)^2 \sigma^2 / 2}\right), \quad (8)$$

the average over the Gaussian ensemble of cycle times with exact controlled evolution and E_{obj} on resonance; the factor $q_k^M \geq 2^{-M}$ makes 2^{-M} a lower bound on the suppression of each excited component, and a sampled average over a finite number of time sets, as in the circuit-level studies below, scatters around Eq. (8) with a relative standard error that grows with M. The width is set by the smallest excitation that carries weight in the input, $\sigma = 5/\Delta_{\text{eff}}$ (Sec. 4.2); choosing it from the bare sector gap would lengthen the controlled evolutions by a factor of up to three (sixteen sites) for no gain.

3.5 The comparators

Four inputs that need no coarse model are fed to the same filter. The *free-fermion determinant* (the N_σ lowest orbitals of the hopping matrix) is prepared by a Givens network of $N(N-1)/2$ rotations per spin. The *ordinary adiabatic ramp* switches on the interaction from it, $H_R(s) = K + \sin^2(\pi s/2)\, U \sum_i n_{i\uparrow} n_{i\downarrow}$; its product-formula version is the fine-chain step with a rescaled interaction angle, with no shift and no coarse term — the analog of the Hartree–Fock ramp of paper I and the baseline every refinement path must beat. The *unitary coupled-cluster singles-and-doubles* (UCCSD) state is the disentangled product $\prod_\mu e^{\theta_\mu(\tau_\mu - \tau_\mu^\dagger)} |\text{HF}\rangle$ of the spin-resolved singles and doubles from the determinant, in a fixed order (all singles, then the doubles, each class in lexicographic order; a different ordering is a different ansatz), optimized by L-BFGS with the analytic gradient; the simulation is exact for the ansatz, every generator being a Givens-type rotation $1 + \sin\theta\, K + (\cos\theta - 1)\Pi$ in the orbital occupation basis. Since most amplitudes are small we also report the *pruned* ansatz obtained by keeping the operators whose converged amplitude exceeds 10^{-3} and re-optimizing. And *ADAPT-VQE* grows the ansatz from the same pool one operator at a time, appending at every iteration the generator with the largest energy gradient (repeats allowed) and re-optimizing all amplitudes warm-started from the previous ones, so that every prefix of the sequence is itself a converged state; this is the adaptive construction that the pruned ansatz approximates. On a device the excitations act on the site register after the Givens network, and we count them with the Pauli-rotation primitive of Sec. 5: a single compiles into two strings, a double into eight, a weight-w string costs $2(w-1)$ CNOTs.

4 Sector-exact results

4.1 The chain refined from four to eight sites

Table 1 and Fig. 1 summarize the refinement of the four-site chain with $N_\uparrow = N_\downarrow = 2$ (36 states) to the eight-site chain (784 states) for $U/t = 0, 2, 4, 8$. Three features stand out. (i) The prolonged state is a good start at every coupling, $p_0 = 0.75$–0.81, with a weak maximum at $U = 2t$; unlike the pairing model, the free-fermion case is not trivial, because the bonding modes are not the lowest fine orbitals. (ii) The sector gap of the fine Hamiltonian falls by a factor of four over the range of U, from $0.53t$ to $0.14t$, but the refinement time does not move: with $\mu_{\mathrm{buf}} = 2t$ the overlap reaches $p > 0.997$ at $T = 8/t$, $1 - p < 10^{-3}$ at $T^* = 10/t$ and $p > 0.9999$ at $T = 16/t$ for all four couplings, and with $\mu_{\mathrm{buf}} = t$ at roughly twice the time. The minimum of the sector gap along Path A sits at $s = 1$ (it is the fine-sector gap itself) in every case, so the global path gap cannot be what controls the transport. (iii) Path B refines at a comparable rate but saturates at the finite-Λ ceiling: at $T = 64/t$, $1 - p = 4.8$–7.3×10^{-3} for $\Lambda = 8t$ and 1.7–3.7×10^{-3} for $\Lambda = 16t$, consistent with the $(t/\Lambda)^2$ admixture (the prolonged state's overlap with the exact ground state of $H_B(0)$ is 0.994–0.995 at $\Lambda = 8t$ and 0.998–0.999 at $16t$). The ceiling, not the transport, limits Path B.

4.2 Which gap controls the refinement: symmetry protection

Both terms of Eq. (6) commute with the total spin and with the point group of the fine lattice (the block structure respects the reflection of the chain and the C_{4v} group of the square), and the prolongation maps irreducible representations onto themselves. The transported state therefore never leaves the symmetry sector of the coarse ground state — a spin singlet, even under reflection, for Table 1 — and the states that can be excited along the path are only those of the same sector. Figure 2 makes this explicit: for $U = 4t$ the lowest fine excitations at $0.223t$ (a reflection-odd triplet) and $0.459t$ (an even triplet) carry weights of order 10^{-31}, and the first state with nonzero weight (4.6×10^{-5}), an even singlet, lies at $\Delta_{\mathrm{eff}} = 0.588t$; at $U = 8t$ the corresponding numbers are $0.137t$ and $0.286t$ (triplets, zero weight) and $\Delta_{\mathrm{eff}} = 0.366t$. Δ_{eff} falls with U much more slowly than Δ, which explains the U-independent T^*.

Figure 3 follows the two gaps along Path A: the sector gap and the gap to the lowest state with the same (S, parity) as the tracked state. The latter never approaches the former; its minimum is Δ_{eff} itself, at $s = 1$. This is the isolated-branch condition of the adiabatic theorem verified numerically along the whole path, within the symmetry sector in which the transport actually takes place. The sufficient adiabatic time of Jansen, Ruskai and Seiler [21], evaluated as in paper I once with the sector gap and once with the symmetry-resolved gap, is smaller by a factor 5 to 17 in the second form but remains three to five orders of magnitude above the empirical $T^* = 10/t$ (Supplemental Material [20]); the two numbers answer different questions, a worst-case operator-norm guarantee for every state in the branch against one overlap for one initial state. The first-order response norm of the path, $\chi_u(s)$, whose integral sets the adiabatic leakage at leading order and which paper I obtained in closed form for the pairing model, is largest at the initial endpoint for every coupling, $\chi_u(0) = 0.38$–0.87 for $U/t = 0$–8, and its modest growth with U is in line with the flat T^* (Supplemental Material [20]).

4.3 The self-similar hierarchy: from the filled dimer to sixteen sites

The self-similarity of Eq. (4) allows the chain of refinements to start from the two-site dimer with $N_\uparrow = N_\downarrow = 2$, whose sector contains a single state — both sites doubly occupied — with $E_{\mathrm{low}} = U - 4t$: the analog of the $k = 2$ determinant that seeded the $2 \to 4$ chain of paper I, requiring no classical or variational preprocessing. The step $2 \to 4$ starts from $p_0 = 0.80$ at $U = 0$ down to 0.48 at $U = 8t$, but with a large effective gap ($\Delta_{\mathrm{eff}} = 2.2t$–$0.8t$) it reaches $p > 0.9997$ by $T = 16/t$ at every U. Composing steps with the *refined* four-site state as the input of the $4 \to 8$ step, and the shift referenced to its variational energy, the chain $2 \to 4 \to 8$ at $U = 4t$ with $T = 16/t$ per step reaches $p = 0.99988$, against 0.99994 from the exact coarse input; with $T = 8/t$ in the first step the second saturates at $p = 0.9979$, the overlap of its input, because the admixture of other coarse

Table 1: Refinement of the open Hubbard chain from $L = 4$ to $L = 8$ sites at $N_\uparrow = N_\downarrow = 2$ ($t = 1$; 36 and 784 states). E_{low} is the ground-state energy of the Galerkin coarse model, Eq. (4), E_{high} the ED ground-state energy of the fine chain, p_0 the weight of the prolonged coarse ground state on the fine ground state, Δ the sector gap of the fine Hamiltonian and Δ_{eff} the lowest excitation energy carrying weight in the prolonged state. $p(T)$ is the overlap after refinement along Path A ($\mu_{\mathrm{buf}} = 2t$) and Path B ($\Lambda = 16t$); the last two columns give the ensemble infidelity, Eq. (8), of the rodeo filter with $\sigma = 5/\Delta_{\mathrm{eff}}$ after six cycles, applied to the prolonged state and to the Path-A state refined for $T = 16/t$.

U/t	E_{low}	E_{high}	p_0	Δ	Δ_{eff}	$p_A(8)$	$p_A(16)$	$p_B(16)$	$p_B(32)$	$1-\mathcal{F}_6$, prol.	$1-\mathcal{F}_6$, ref.
0	−6.236068	−6.822948	0.777	0.532	0.879	0.99718	0.99997	0.99660	0.99838	4.5×10^{-3}	5.4×10^{-7}
2	−5.437971	−6.032090	0.814	0.319	0.780	0.99876	0.99996	0.99594	0.99849	3.6×10^{-3}	5.9×10^{-7}
4	−4.976573	−5.634427	0.797	0.223	0.588	0.99837	0.99994	0.99316	0.99769	4.0×10^{-3}	9.2×10^{-7}
8	−4.558586	−5.277973	0.750	0.137	0.366	0.99761	0.99991	0.99205	0.99731	5.2×10^{-3}	1.4×10^{-6}

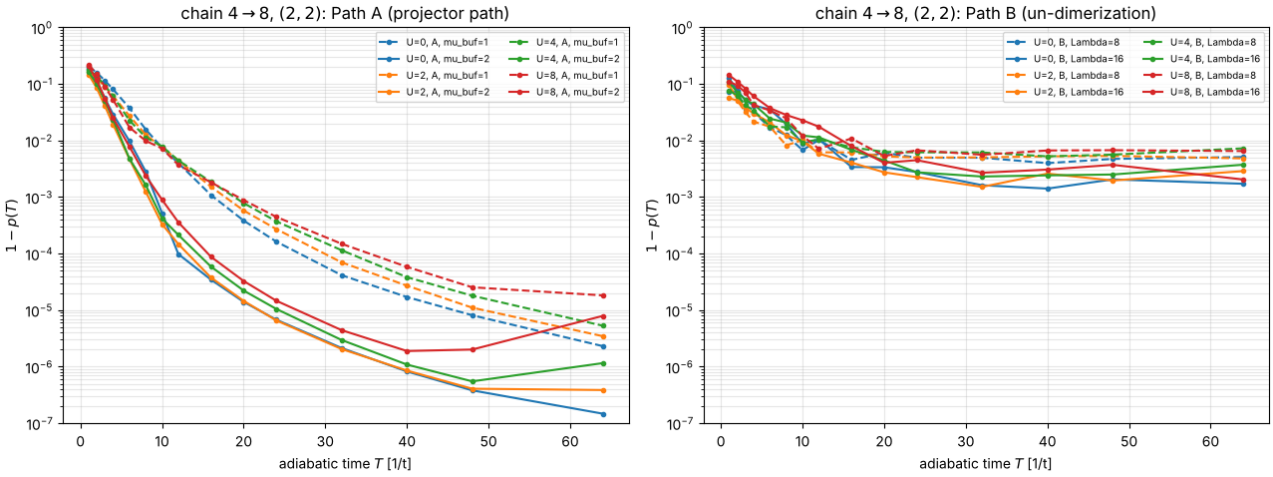


Figure 1: Refinement of the open chain $4 \to 8$ at $(2,2)$: $1-p(T)$ along Path A (left; solid $\mu_{\mathrm{buf}} = 2t$, dashed $\mu_{\mathrm{buf}} = t$) and along Path B (right; solid $\Lambda = 16t$, dashed $\Lambda = 8t$) for $U/t = 0, 2, 4, 8$. Path A reaches $1-p < 10^{-4}$ by $T = 16/t$ at every coupling; Path B saturates at the finite-Λ ceiling of a few 10^{-3}.

states refines to other fine states. The composed chain also beats a single-shot refinement $2 \to 8$ with blocks of four ($p_0 = 0.579$, $p = 0.9966$ at $T = 16/t$). The protection survives a doubling of the system: refining the eight-site chain to sixteen sites at $(2,2)$ ($784 \to 14\,400$ states, $U = 4t$) starts from $p_0 = 0.869$, faces a fine-sector gap of only $0.051t$, and still reaches $p = 0.9924$ at $T = 8/t$, 0.99931 at $T = 16/t$ and 0.99995 at $T = 32/t$, because the first state to carry weight lies at $\Delta_{\mathrm{eff}} = 0.138t$ while the eight other states below $0.4t$, including the one that defines the sector gap, carry none. Three steps from the filled dimer, $2 \to 4 \to 8 \to 16$ with $T = 16/t$ each and the refined state as the input of every step, give $p = 0.99923$ for the sixteen-site ground state. The tables and figures of the dimer step and of the composed chains are in the Supplemental Material [20].

4.4 Two dimensions: the plaquette and the wrong symmetry sector

For the square lattice the first step is $2\times2 \to 4\times4$. With one particle per spin ($16 \to 256$ states) the refinement behaves as in one dimension: both ground states are A_1, $p_0 = 0.80$–0.85 for $U/t = 0, 4, 8$, $\Delta_{\mathrm{eff}} \simeq 2.1t$–$2.2t$, and $p > 0.9997$ at $T = 8/t$. With two particles per spin ($36 \to 14\,400$ states) the prolonged coarse ground state has *zero* overlap with the fine ground state, and the symmetry diagnostics identify why (Fig. 4). The half-filled coarse plaquette (Galerkin parameters $t/2$, $U/4$) has, for every $U > 0$, a B_1 singlet ground state with an A_2 triplet close above it ($0.071t$ at $U = 4t$); the dilute fine lattice, four electrons on sixteen open sites, has an A_2 *triplet* ground state for every $U > 0$ studied ($U/t = 1$–8), the B_1 singlet lying $0.04t$–$0.11t$ above it. The prolongation commutes with $\mathbf{S}^2$ and with C_{4v}, so the two cannot be connected: the fine ground state has

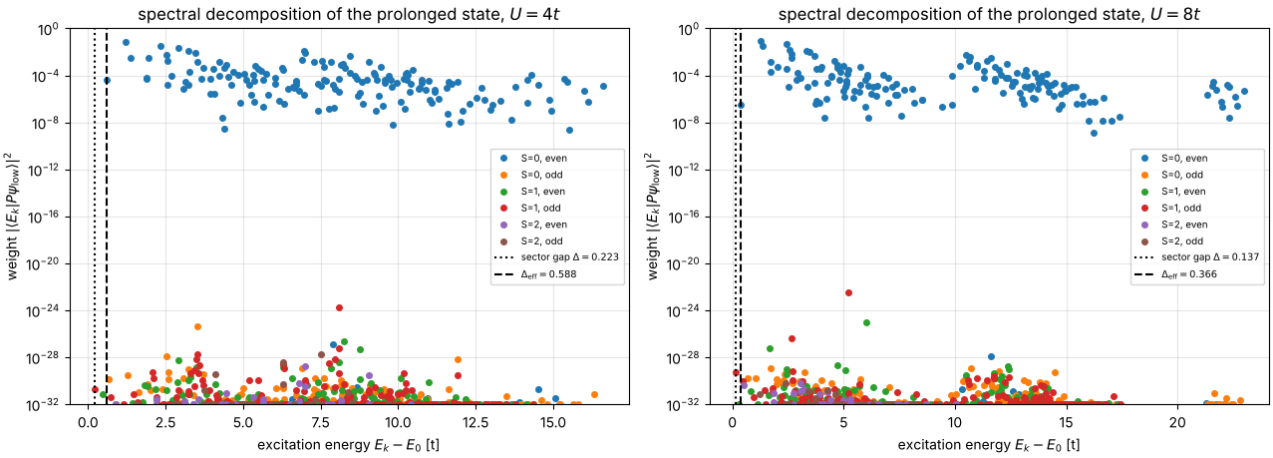


Figure 2: Spectral decomposition of the prolonged coarse ground state over the eigenstates of the eight-site chain at $U = 4t$ (left) and $8t$ (right), each eigenstate labeled by its total spin and reflection parity. Only even singlets carry weight; the triplets and odd states, including the ones that define the sector gap Δ (dotted), sit at round-off level. The lowest state with weight defines $\Delta_{\rm eff}$ (dashed).

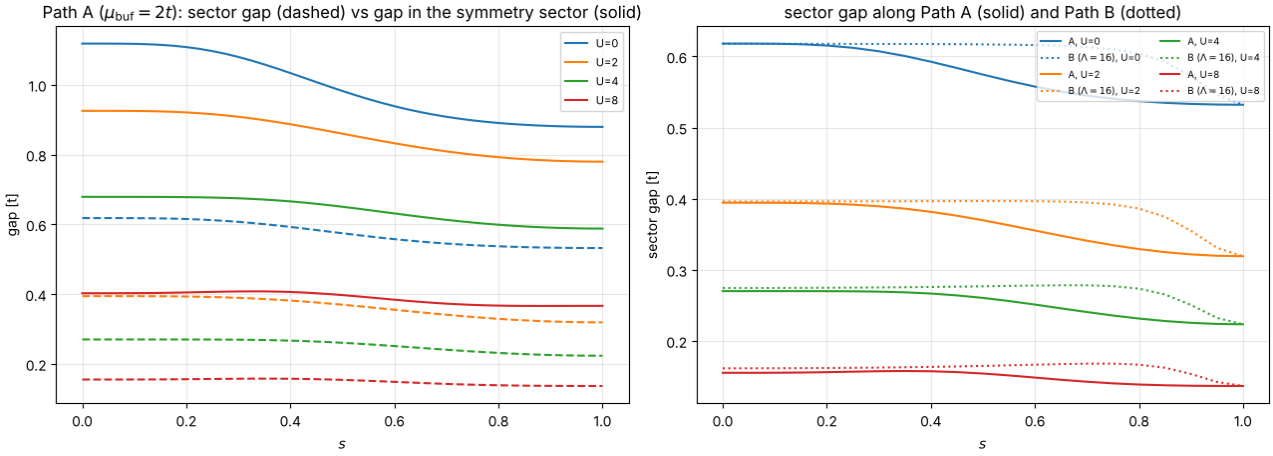


Figure 3: Left: along Path A ($\mu_{\rm buf} = 2t$), the sector gap (dashed) and the gap within the symmetry sector of the tracked state (solid) for $U/t = 0, 2, 4, 8$. Right: the sector gap along Path A (solid) and Path B with $\Lambda = 16t$ (dotted); both reach their minimum, the fine-sector gap, at $s = 1$.

weight 0 on the prolonged coarse ground state, 0.647 on the prolonged A_2 triplet and 0.007 on all other coarse states together ($U = 4t$). Refining that triplet along Path A ($\mu = E_1^{\rm low} + 2t$) reaches $p = 0.99944$ at $T = 8/t$ and 0.99997 at $T = 16/t$ although the fine-sector gap is only $0.090t$, since the lowest state of the $(A_2, S = 1)$ sector that carries weight lies $1.77t$ above the ground state; the same path started from the coarse ground state gives $p = 3 \times 10^{-27}$. The lesson for the method is that the state to refine is the lowest coarse eigenstate *in the symmetry sector of the target*, which need not be the coarse ground state, that the coarse solver must be run with the target's quantum numbers imposed or at least diagnosed, and that the filter cannot rescue a wrong choice, since a state of zero overlap has nothing to amplify. Conversely the refinement is a sensitive detector of a level crossing between the coarse and the fine problem — here a singlet–triplet crossing as the filling goes from one half to one eighth, a property of the filling, not of U, over the whole range studied. The spectra and the table of the plaquette cases are in the Supplemental Material [20].

4.5 Filtering, and the variational inputs

With $\sigma = 5/\Delta_{\rm eff}$ ($5.7/t$ to $13.6/t$ for $U/t = 0$ to 8) the exact ensemble filter, Eq. (8), applied to the prolonged state takes the infidelity from $1 - p_0 \simeq$

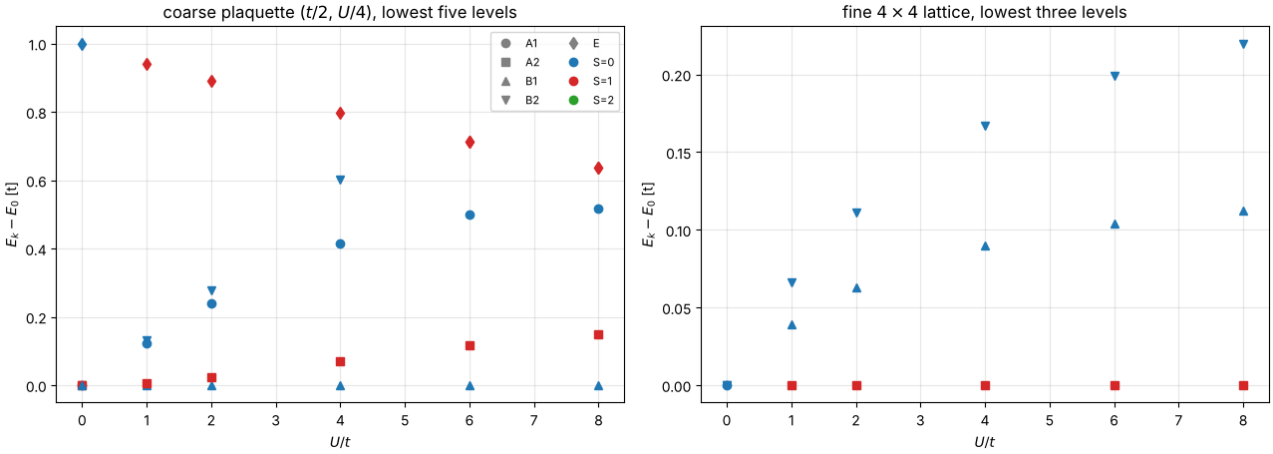


Figure 4: Lowest levels of the coarse plaquette (left, Galerkin parameters $t/2$, $U/4$) and of the fine 4×4 lattice (right) at $(2, 2)$ versus U/t, labeled by C_{4v} irrep (marker) and total spin (color). For every $U > 0$ the coarse ground state is a B_1 singlet and the fine ground state an A_2 triplet; the prolongation conserves both labels.

0.2 to 4–5 $\times 10^{-3}$ after six cycles and 2–3 $\times 10^{-4}$ after ten, with an acceptance that converges to p_0; applied to the Path-A state refined for $T = 16/t$ it starts from 10^{-4} and reaches 0.5–1.4$\times 10^{-6}$ after six cycles; applied to the eight-site UCCSD state it starts from 4.7×10^{-4}, 3.1×10^{-3} and 1.2×10^{-2} at $U/t = 2, 4, 8$ and reaches 7×10^{-6}, 5×10^{-5} and 2×10^{-4} (figure in the Supplemental Material [20]). The refined input converges faster per cycle by the ratio of the initial infidelities; whether the cycles saved pay for the refinement gates is the circuit-level question of Sec. 5.

The variational tables are in the Supplemental Material [20]; what matters here is three numbers at $U = 4t$. The 198-parameter single-layer UCCSD on the eight-site chain reaches $1 - F = 3.1 \times 10^{-3}$ at 3.1×10^4 CNOTs; 98 of its amplitudes exceed 10^{-3}, and keeping those (12 singles, 86 doubles) and re-optimizing gives the same accuracy, 3.145×10^{-3} against 3.143×10^{-3}, at 1.5×10^4 CNOTs. Used as the coarse solver on the four-site chain, the 26-parameter ansatz ($1 - F = 3.3 \times 10^{-3}$) prolongs with the same p_0 as the exact coarse state (0.800 against 0.797) but the refinement then saturates at $p = 0.9967$ instead of 0.9999: the coarse stage must be as accurate as the target, or the filter must do the rest.

5 Circuits and the matched comparison at eight sites

5.1 Pauli structure and step costs

In the block Jordan–Wigner ordering an open chain of L_f sites on $2L_f$ qubits has $4(L_f - 1)$ hopping strings $\frac{t}{2}(X_j X_{j+1} + Y_j Y_{j+1})$ of weight two, L_f interaction strings $\frac{U}{4} Z_j Z_{j+L_f}$ and $2L_f$ single-Z terms: 52 strings of weight ≤ 2 for $L_f = 8$, verified against the full-space Pauli decomposition on eight qubits, against the 325 strings of weight up to eight of the $k = 4$ pairing model on half the qubits. With the Pauli-rotation primitive of paper I — $2(w - 1)$ CNOTs per weight-w string, two more per rotation when controlled — a second-order Trotter step of the eight-site chain costs 144 CNOTs and a controlled one 352; at sixteen sites, 304 and 736. The product formula groups the bonds into two layers of disjoint bonds plus the diagonal interaction; each layer is a product of commuting two-qubit blocks and is realized exactly, so the Trotter error comes only from the non-commutativity between the three groups, as on hardware. The prolongation $4 \to 8$ costs 16 CNOTs. The embedded coarse term of Path A is the coarse Hubbard step on the bonding register (24 rotations) with every rotation conditioned on the eight antibonding qubits being empty: with a clean-ancilla ladder of relative-phase Toffolis ($6(c - 1) + 2$ extra CNOTs per rotation) it costs 1 088 CNOTs per application, 7.6 fine steps, and at sixteen sites (16 controls, 48 rotations) 4 856, sixteen fine steps. Either way the coarse term,

not the fine step, is the cost of Path A.

5.2 Trotterized refinement and the baselines

With second-order product formulas over the groups $[P(H_{\rm low}-\mu)P^\dagger$ split into its three conditioned coarse groups, $K_{\rm even}, K_{\rm odd}, D-\mu]$ for Path A and $[(1+\Lambda\cos^2)K_{\rm even}, K_{\rm odd}, D]$ for Path B, the exact-slice overlap at $T=16/t$, $U=4t$ is reproduced to within 10^{-3} with $n_s=128$ steps for Path A ($p=0.99982$), 256 for Path B at $\Lambda=8t$ (0.99213) and 512 at $\Lambda=16t$, against 43 steps for the pairing path of paper I: the shifted generators have larger norms ($\|B-A\|=14$ for Path A, $1+\Lambda$ on the intra-block bonds for Path B), which makes $\Lambda=8t$ the better operating point of the two-local path despite its higher ceiling. At these step counts the refinement costs 2.97×10^5 CNOTs for Path A (2.8×10^5 in the coarse term) and 3.7×10^4 for Path B. The ramp from the free-fermion determinant (Givens network 112 CNOTs) has $p_0=0.7804$ — essentially the overlap of the prolonged state, 0.7974 — and the same $\Delta_{\rm eff}=0.588t$, since it lies in the same symmetry sector; with exact slices it reaches $p=0.99686$, 0.99960 and 0.99989 at $T=8,12,16$, as fast as Path A and far above the ceiling of Path B, and its product-formula version needs 64 steps at $T=8/t$ (9 328 CNOTs) and 128 at $T=16/t$ (0.999659, 18 544 CNOTs). The step scans are in the Supplemental Material [20].

5.3 Matched-observable comparison

Following Sec. V.C of paper I we feed the same Trotterized rodeo filter — second order, fixed step δt, $\sigma=5/\Delta_{\rm eff}=8.5/t$, forty Gaussian time sets drawn once and shared by all inputs, up to six cycles — to eight inputs: the prolonged coarse ground state, the free-fermion determinant, the pruned and the full UCCSD states, the ramp states at $T=8/t$ and $16/t$, the Path-B state ($\Lambda=8t$, $n_s=256$) and the Path-A state ($n_s=128$), and compare the acceptance-weighted ensemble infidelity of the accepted runs against the CNOTs per attempt and per accepted preparation (input plus filter, early termination after the first failed ancilla measurement, the expectation over the time sets). Two implementation details are forced by the Hubbard norms. The fixed step is diagonalized once, so that n steps are applied exactly and the cycle times are quantized to multiples of δt (on average 68 steps per cycle, 2.4×10^4 CNOTs); the filter then projects on the eigenvector of the step unitary closest to the ground state, whose eigenphase is what the energy scan of the rodeo locates on a device, and we take $E_{\rm obj}$ equal to it rather than to E_0. The Trotter floor of the filter is that eigenvector's infidelity with the exact ground state: 2.6×10^{-3} at $\delta t=0.25/t$ (the step of paper I, where $E_{\rm obj}-E_0=0.09t$ would also have spoiled the acceptance), 5.8×10^{-5} at $\delta t=0.1/t$ and 3.5×10^{-6} at $0.05/t$; we operate at $\delta t=0.1/t$, where $E_{\rm obj}-E_0=0.013t$. Because the cycle times are quantized every cycle shares the eigenvectors of the step unitary, so the long-run state of the filter is that eigenvector itself and, with $E_{\rm obj}$ on its eigenphase, the acceptance tends to the input's weight on it instead of decaying — the fixed-step analog of the averaged post-selection map of paper I. The sampled infidelities carry bootstrap relative standard errors of up to 18% for the unrefined inputs at $M=6$ and below 10% for the refined and variational ones.

Table 2 and Fig. 5 give the verdict at eight sites; the operating points (minimum cost at each target, with the cycle count that attains it) are tabulated in the Supplemental Material [20]. At every target the cheapest route is the ramp from the free-fermion determinant at $T=16/t$: 1.9×10^4 CNOTs for 10^{-3} without any filter cycle, 4.0×10^4 for 3×10^{-4} with one cycle and 9.1×10^4 for 10^{-4} with three. The pruned UCCSD state costs 3.3 and 2.9 times more at the first two targets and does not reach 10^{-4} within six cycles; the full pool is a further 10–25% more expensive. Path B at $\Lambda=8t$ costs 7.4 and 4 times the ramp, because its preparation is twice as expensive and starts twenty times worse owing to the ceiling. Path A delivers the best unfiltered input ($1-p=1.8\times10^{-4}$) but at 3×10^5 CNOTs, sixteen times the ramp; it is the only input other than the ramp to reach 10^{-4}, at 3.4×10^5 CNOTs. The prolonged state and the bare determinant need six cycles and 1.5–1.6×10^5 CNOTs to reach 5×10^{-3} and reach none of the targets. Only the inputs that start within a decade of the floor converge to it within six cycles; the floor is set by δt, but whether it is reached is set by the input. The costs are per accepted preparation at the operating point of each input; the setup costs — the classical optimization and its measurements for the variational states, the calibra-

Table 2: Matched-observable comparison for the chain $4 \to 8$ at $U = 4t$: the same Trotterized rodeo filter (second order, $\delta t = 0.1/t$, $\sigma = 8.5/t$, E_{obj} the Trotter-effective ground energy, forty time sets) applied to every input. p is the overlap of the input and "prep" its CNOT count (prolongation 16; Givens network 112; refinement or ramp n_s second-order steps of 144 CNOTs, plus $2 \times 1\,088$ per step for the coarse term of Path A; UCCSD pruned 15 120, full 30 864); $M = 0$ is the unfiltered input. For $M = 2$ and 6 cycles: the ensemble infidelity (bootstrap relative standard error in parentheses), the cumulative acceptance P, and the CNOTs per accepted preparation (input plus filter, early termination; the CNOTs per attempt are P times smaller).

input	p	prep	$M = 0$	$M = 2$			$M = 6$		
			$1 - \mathcal{F}$	$1 - \mathcal{F}$	P	CNOTs	$1 - \mathcal{F}$	P	CNOTs
prolonged	0.7974	16	2.0×10^{-1}	6.0×10^{-2} (7%)	0.848	5.2×10^{4}	4.7×10^{-3} (18%)	0.802	1.5×10^{5}
determinant	0.7804	112	2.2×10^{-1}	6.6×10^{-2} (5%)	0.831	5.3×10^{4}	5.0×10^{-3} (17%)	0.779	1.6×10^{5}
UCCSD pruned (98)	0.9969	15 120	3.1×10^{-3}	8.2×10^{-4} (6%)	0.997	6.2×10^{4}	1.1×10^{-4} (7%)	0.997	1.6×10^{5}
UCCSD full (198)	0.9969	30 864	3.1×10^{-3}	8.1×10^{-4} (6%)	0.997	7.8×10^{4}	1.1×10^{-4} (7%)	0.997	1.7×10^{5}
ramp, $T = 8$, $n_s = 64$	0.9969	9 328	3.1×10^{-3}	7.8×10^{-4} (14%)	0.998	5.6×10^{4}	8.9×10^{-5} (10%)	0.997	1.5×10^{5}
ramp, $T = 16$, $n_s = 128$	0.9997	18 544	3.4×10^{-4}	1.3×10^{-4} (6%)	1.000	6.5×10^{4}	6.2×10^{-5} (3%)	1.000	1.6×10^{5}
Path B ($\Lambda = 8t$), $n_s = 256$	0.9921	36 880	7.9×10^{-3}	2.0×10^{-3} (6%)	0.994	8.4×10^{4}	2.0×10^{-4} (9%)	0.992	1.8×10^{5}
Path A (split), $n_s = 128$	0.9998	296 976	1.8×10^{-4}	9.1×10^{-5} (2%)	1.000	3.4×10^{5}	6.0×10^{-5} (1%)	1.000	4.4×10^{5}

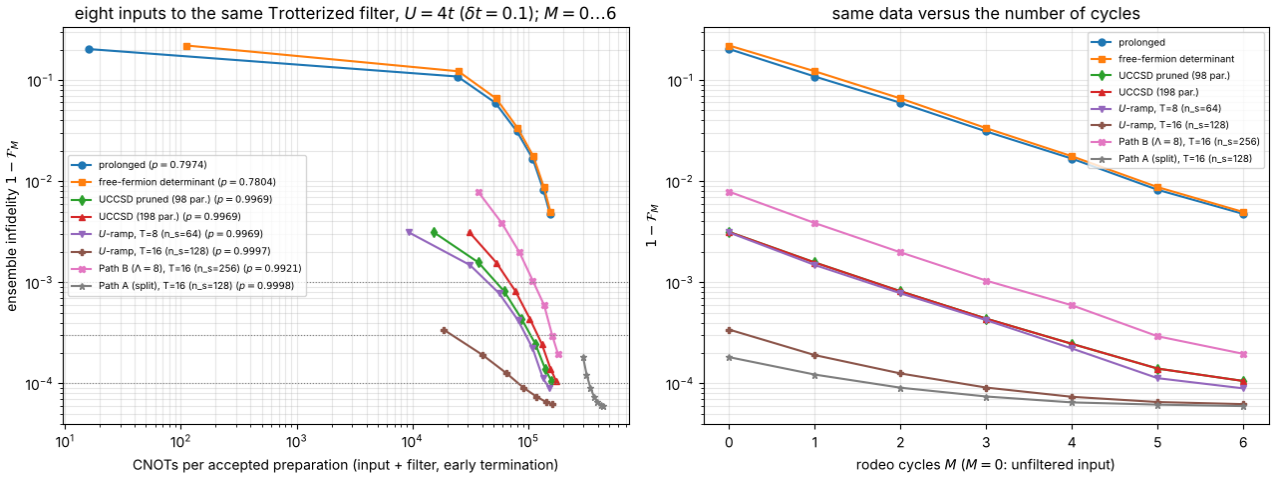


Figure 5: Cost versus accuracy for the chain $4 \to 8$ at $U = 4t$: ensemble infidelity against CNOTs per accepted preparation (left; the first point of each curve, $M = 0$, is the unfiltered input, dotted lines the targets 10^{-3}, 3×10^{-4}, 10^{-4}) and against the number of cycles (right) for the eight inputs of Table 2, all cleaned by the same Trotterized filter. The refined inputs reach the Trotter floor of 5.8×10^{-5} within six cycles; the unrefined ones do not.

tion of $(T, \mu_{\mathrm{buf}}, n_s)$ for the adiabatic inputs, the energy scan that locates E_{obj} — are not included and are amortized over the preparations that use them, and the refinement parameters are the first grid values that meet the accuracy criterion, not optimized ones; a joint search may move every curve by a factor of order unity, but not the factor sixteen of the conditioned coarse term nor the twenty-fold ceiling of Path B.

6 Eight to sixteen sites at the circuit level

The comparison of Sec. 5.3 is at one size, where the free-fermion determinant has the same overlap and the same Δ_{eff} as the prolonged coarse eigenstate; the case for refinement rests on the regime where the coarse eigenstate is the better start. Here the chain $8 \to 16$ at $(2, 2)$, $U = 4t$ ($784 \to 14\,400$ states, thirty-two qubits) is taken to the circuit level with the same accounting. Every product-formula factor of the fine chain is applied through the Kronecker structure of Eq. (2) as a pair of 120×120 exponentials, the embedded coarse term of Path A exactly through the 784-dimensional coarse space, and the variational comparators with sparse generators; the Trotter-effective ground state of the fixed-step filter, which cannot be obtained by diagonalizing the 14 400-dimensional step unitary, is obtained by iterating the quantized-time filter itself from the exact ground state (details and val-

idation against the dense eight-site machinery in the Supplemental Material [20]).

The inputs. The prolonged coarse ground state now starts ahead of the determinant, $p_0 = 0.869$ against 0.716, with the same $\Delta_{\text{eff}} = 0.138t$ for both (the determinant lies in the same symmetry sector), while the sector gap is $0.051t$ and the sector gap along the ramp falls from $0.165t$ to $0.051t$. Figure 6(left) shows the exact-slice transports. The ramp needs $T = 32/t$ for $1-p = 4.1\times 10^{-3}$ and $T = 64/t$ for 8.1×10^{-4} (1.1×10^{-3} at $48/t$), against $T = 8/t$ for $p = 0.9924$ ($T^* = 16/t$, $p = 0.99931$; 0.99995 at $32/t$) along Path A with $\mu_{\text{buf}} = 2t$; a larger shift does not help at this size ($p(16) = 0.99879$ and 0.99792 at $\mu_{\text{buf}} = 4t$ and $8t$), because it raises the norm of the generator without widening a gap that is already set by Δ_{eff}. Path B sits at its ceiling from $T = 16/t$ on, $p = 0.99492$ and 0.99531 at $T = 16/t$ and $32/t$ for $\Lambda = 8t$ (0.99500 and 0.99605 for $\Lambda = 16t$), the same few 10^{-3} as at eight sites. The product formulas need $n_s = 128$ second-order steps for the ramp at $T = 16/t$ ($p = 0.96927$, 39 392 CNOTs) and 256 at $T = 32/t$ (0.99556, 78 304 CNOTs; with 32 steps, $\delta t = 1/t$, the ramp overlap collapses to 0.005), 256 for Path B at $\Lambda = 8t$, $T = 16/t$ (0.99476, 77 856 CNOTs plus the coarse preparation); Path A at $T = 16/t$ with the coarse term split into its conditioned groups reaches $p = 0.99835$ at $n_s = 128$ for 1.28×10^6 CNOTs, of which 1.24×10^6 in the coarse term, while at $T = 8/t$ the product formula plateaus at $p = 0.99079$ from $n_s = 128$ on, 1.6×10^{-3} below the exact-slice value, so that the shorter path cannot be made to reproduce its exact transport at any step count. The single-layer UCCSD ansatz on sixteen sites has 1 022 parameters, costs 3.4×10^5 CNOTs and reaches only $1-F = 7.0\times 10^{-3}$ (against 3.1×10^{-3} with 198 parameters at eight sites); pruning to the 255 amplitudes above 10^{-3} keeps that accuracy at 8.8×10^4 CNOTs. Figure 6(right) shows ADAPT-VQE with the same pool: at eight sites the adaptive ansatz passes the single-layer accuracy at 66 operators and reaches 10^{-3} at 149 and 3×10^{-4} at 194 operators (a third of the template's operators, chosen adaptively and with repeats, do what the full template does); at sixteen sites it needs 187 operators (6.8×10^4 CNOTs) to reach 10^{-2} and has not reached 3×10^{-3} after 200. The convergence is slow because the ground state spreads over many small amplitudes rather than a few large ones, the opposite of the pairing case of paper I, where five operators sufficed at eight qubits; and, unlike there, the adaptive ansatz does not undercut the ramp even at eight sites, where the 149-operator state (2.3×10^4 CNOTs) fed to the filter of Sec. 5.3 reaches 10^{-3}, 3×10^{-4} and 10^{-4} at 2.3×10^4, 6.9×10^4 and 1.5×10^5 CNOTs against the ramp's 1.9×10^4, 4.0×10^4 and 9.1×10^4.

The filter at thirty-two qubits. With $\sigma = 5/\Delta_{\text{eff}} = 36/t$ and $\delta t = 0.1/t$ a cycle takes 289 steps on average, 2.1×10^5 CNOTs, nine times the cycle of Sec. 5.3: 4.3 times more steps, since Δ_{eff} is 4.3 times smaller, and 2.1 times more per step. The Trotter-effective ground energy lies $4.9\times 10^{-3}t$ above E_0 and the floor of the filter is 3.7×10^{-5}. The cost of the filter therefore scales as $1/\Delta_{\text{eff}}$ per cycle at fixed δt, and an input that is cheap but needs several cycles pays this price in full at every size.

Matched comparison. Table 3 and Fig. 7 feed eight inputs to this filter with twenty shared time sets: the determinant, the prolonged coarse ground state, the pruned UCCSD state, the ramp at $T = 16/t$ and $32/t$, Path B ($\Lambda = 8t$, $T = 16/t$), Path A ($T = 16/t$) and the ADAPT state at its first iteration below 10^{-2}; every input that starts from the coarse eigenstate is charged, in addition, its preparation by the cheapest eight-site route of Table 2 that reaches 3×10^{-4}, the ramp at $T = 16/t$ (18 544 CNOTs). The floor is reached by none of the inputs within six cycles (the best, Path A, is at 7.4×10^{-5}, twice the floor); the bootstrap relative standard errors reach 20–35% at $M = 6$ with twenty time sets, and the operating points below are read with that uncertainty. The ramp at $T = 32/t$ is the cheapest route at every target: 7.8×10^4 CNOTs for 10^{-2} without a cycle, 3.0×10^5 for 3×10^{-3} with one, 7.9×10^5 for 10^{-3} with three and 1.3×10^6 for 3×10^{-4} with five. Path B is within 1–6% of it at the three tighter targets, a tie within the sampling error, and 23% above it at 10^{-2}, where only the preparation counts: its preparation (9.6×10^4 CNOTs including the coarse eigenstate) and its starting infidelity (5.2×10^{-3} against 4.4×10^{-3}) match the ramp's, so at this size the coarse eigenstate transported along the cheapest path is as good a start

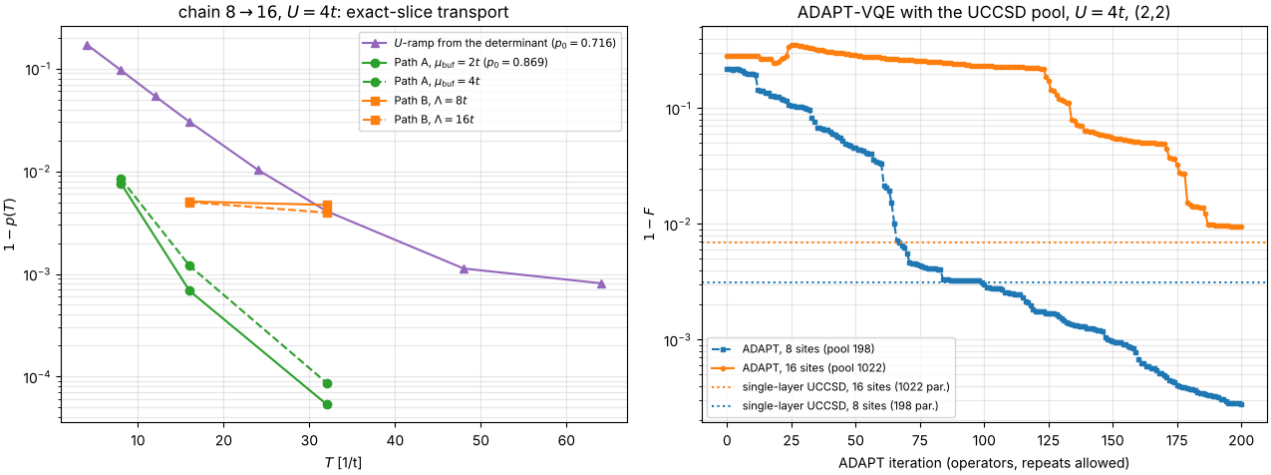


Figure 6: The chain $8 \to 16$ at $(2,2)$, $U = 4t$. Left: $1 - p(T)$ with exact slices for the ramp from the free-fermion determinant, Path A with $\mu_{\mathrm{buf}} = 2t$ and $4t$ and Path B with $\Lambda = 8t$ and $16t$. Right: ADAPT-VQE with the UCCSD pool at eight and at sixteen sites, infidelity against the number of operators, with the single-layer UCCSD values as horizontal lines.

as the determinant transported for twice the time. The variational routes cost 1.3–1.9 times more at 3×10^{-3} and 10^{-3} — the adaptive state at 10^{-2} is the cheapest unfiltered input (6.8×10^4 CNOTs) but then needs four cycles for 10^{-3} — and come within 1–13% of the ramp only at 3×10^{-4}, where every input needs five or six cycles and the preparation no longer matters. Path A delivers the best unfiltered input, 1.7×10^{-3}, and reaches 10^{-3} with one cycle and 3×10^{-4} with three, at 1.9 and 1.6 times the cost of the ramp: the factor sixteen of eight sites has become a factor two, because the ramp itself now needs three cycles, 6.4×10^5 CNOTs of filtering, before it reaches 10^{-3}, while the 1.2×10^6 CNOTs of the conditioned coarse term buy a start that needs one. The ramp at $T = 16/t$, the cheapest input at eight sites, needs six cycles for 10^{-3} here and is beaten by the two-local path and by both variational inputs, and the prolonged state and the determinant reach 2.4×10^{-3} and 7.1×10^{-3} after six cycles, with acceptances 0.87 and 0.72.

7 Discussion

What the benchmark shows. At sixteen qubits the ordinary adiabatic ramp from the free-fermion determinant is the cheapest route to every target accuracy, by a factor of three over the pruned variational ansatz, four to seven over the two-local refinement path and sixteen over the projector path, for a reason specific to the size: the determinant has the same overlap and the same symmetry-protected gap as the prolonged coarse eigenstate, so the ramp starts from the same place, travels at the same speed and pays only for fine-chain steps. At thirty-two qubits the premise changes — the coarse eigenstate starts ahead, the ramp needs twice the time, the single-layer ansatz has lost two orders of magnitude and the adaptive ansatz barely reaches 10^{-2} with two hundred operators — and the margins collapse: the two-local refinement path ties with the ramp from 3×10^{-3} down, the variational routes cost 1.3–1.9 times more, and the projector path costs twice the ramp instead of sixteen times. The ramp is still the cheapest, but the comparison is now decided by the filter cycles each input needs, at 2×10^5 CNOTs per cycle, not by the preparation — and the input that needs fewest is the refined one. The trend from eight to sixteen sites is therefore the one the refinement proposal predicted: as the free-fermion overlap decays, the coarse eigenstate becomes the better start, and the open question is no longer whether the refinement stage can pay but whether its transport can be compiled without the conditioned coarse term — Path B, whose ceiling of a few 10^{-3} is what keeps it from inheriting the 1.7×10^{-3} start of Path A at a fifteenth of the cost. Two lessons are inherited from paper I and sharpened. A conditioned coarse term is never affordable: at sixteen sites it costs sixteen fine steps per application, and the two-local path or a schedule that

Table 3: Matched comparison for the chain $8 \to 16$ at $U = 4t$ (thirty-two qubits): the same Trotterized rodeo filter (second order, $\delta t = 0.1/t$, $\sigma = 36/t$, $E_{\rm obj}$ the Trotter-effective ground energy, twenty time sets) applied to every input. "prep" counts the Givens network (480 CNOTs), the prolongation (32), the second-order steps (304 CNOTs, plus $2 \times 4\,856$ per step for the coarse term of Path A), the variational circuits, and, for every input that starts from the coarse eigenstate, its preparation by the cheapest eight-site route of Table 2 that reaches 3×10^{-4} (18 544 CNOTs). $1 - \mathcal{F}$ at $M = 0$ (the unfiltered input) and after six cycles (bootstrap relative standard error in parentheses; the cumulative acceptance after six cycles equals p to two digits for every input); then the minimum CNOTs per accepted preparation at each target infidelity, with the cycle count in parentheses (a dash: not reached within six cycles). The full $M = 0$–6 data are in the Supplemental Material [20].

input	p	prep	$1-\mathcal{F}_0$	$1-\mathcal{F}_6$	10^{-2}	$3{\times}10^{-3}$	10^{-3}	$3{\times}10^{-4}$
determinant	0.7162	480	$2.8{\times}10^{-1}$	$7.1{\times}10^{-3}$ (18%)	$1.5{\times}10^{6}$ (6)	—	—	—
prolonged	0.8692	18 576	$1.3{\times}10^{-1}$	$2.4{\times}10^{-3}$ (13%)	$1.3{\times}10^{6}$ (5)	$1.4{\times}10^{6}$ (6)	—	—
UCCSD pruned (255)	0.9930	87 776	$7.0{\times}10^{-3}$	$1.3{\times}10^{-4}$ (12%)	$8.8{\times}10^{4}$ (0)	$5.6{\times}10^{5}$ (2)	$1.1{\times}10^{6}$ (4)	$1.3{\times}10^{6}$ (5)
ADAPT (187 op.)	0.9901	68 008	$9.9{\times}10^{-3}$	$1.7{\times}10^{-4}$ (11%)	$6.8{\times}10^{4}$ (0)	$5.4{\times}10^{5}$ (2)	$1.0{\times}10^{6}$ (4)	$1.4{\times}10^{6}$ (6)
ramp, T=16, n_s=128	0.9693	39 392	$3.1{\times}10^{-2}$	$4.2{\times}10^{-4}$ (35%)	$5.2{\times}10^{5}$ (2)	$1.0{\times}10^{6}$ (4)	$1.4{\times}10^{6}$ (6)	—
ramp, T=32, n_s=256	0.9956	78 304	$4.4{\times}10^{-3}$	$8.6{\times}10^{-5}$ (23%)	$7.8{\times}10^{4}$ (0)	$3.0{\times}10^{5}$ (1)	$7.9{\times}10^{5}$ (3)	$1.3{\times}10^{6}$ (5)
Path B (Λ=8t), T=16, n_s=256	0.9948	96 400	$5.2{\times}10^{-3}$	$1.4{\times}10^{-4}$ (13%)	$9.6{\times}10^{4}$ (0)	$3.2{\times}10^{5}$ (1)	$8.1{\times}10^{5}$ (3)	$1.3{\times}10^{6}$ (5)
Path A (split), T=16, n_s=128	0.9983	1 300 624	$1.7{\times}10^{-3}$	$7.4{\times}10^{-5}$ (17%)	$1.3{\times}10^{6}$ (0)	$1.3{\times}10^{6}$ (0)	$1.5{\times}10^{6}$ (1)	$2.0{\times}10^{6}$ (3)

avoids the embedding altogether is the only form of the projector path worth compiling. And the figure of merit is the cost at fixed accuracy per accepted preparation with the filter cycles included, where the cycle cost grows as $1/\Delta_{\rm eff}$: at fixed cycle count every input "converges to the floor" only in the sense that the best ones do.

Which gap matters. The quantity that controls the refinement is the symmetry-protected effective gap $\Delta_{\rm eff}$, not the sector gap; this is why T^* is flat in U and in size, why the sufficient time of the adiabatic theorem is a closer (though still remote) estimate when evaluated with the symmetry-resolved gap, and why $\Delta_{\rm eff}$ is the right width for the filter. Paper I found that its scan of the refinement time against the shift-controlled gap did not identify a power law; the lesson here is that any such scan must be made against the gap in the symmetry sector of the tracked state, with the response norm held fixed, before an exponent can be read off. The zero-overlap plaquette is the converse and a warning for any refinement, fusion or nucleation scheme: the coarse state must carry the quantum numbers of the target, which the ED diagnostics of Sec. 2 — or, on a device, measurements of $\mathbf{S}^2$ and of the point-group characters — can check before any gate is spent.

Lattice versus basis refinement, and coarse accuracy. The lattice refinement is structurally cleaner than the basis refinement of paper I: a Galerkin coarse model in closed form, a self-similar hierarchy that starts from a single basis state, a two-local Hamiltonian with $O(L_f)$ Pauli strings of weight two, a prolongation of $O(N_c)$ gates. Against this stand the conditioned coarse term, Path B as its two-local replacement with a $(t/\Lambda)^2$ ceiling and a step count that grows with Λ, and three to ten times more Trotter steps than the pairing path for the same adiabatic time, because the shifted generators have larger norms. Composing steps with refined inputs and starting from a variational coarse state show the same ceiling: an admixture of other coarse eigenstates refines to other fine eigenstates and caps the final overlap at roughly the input fidelity, so the coarse stage must be as accurate as the target, or the filter must pay for the difference; the filled dimer, which is exact, is from this point of view the best coarse state the hierarchy has.

Outlook. At fixed U/t the hierarchy is self-similar, so refinement does not by itself reach strong coupling; a path that raises U with the resolution, or a refinement at fixed filling, are the options. In one dimension the dimer-seeded hierarchy and the nucleation of a chain from a dimer [17] start from the same state and reach the same target by refining resolution versus growing the domain, and the Hubbard chain is the natural place to compare the two at matched gate count; the optimized variants of either (a warm-started variational path) are the obvious way to cut the 10^2–10^3 Trotter steps found here. The refinement and filter parameters are calibrated against the exact answer; a device-level protocol (an energy scan for $E_{\rm obj}$, acceptance statistics for

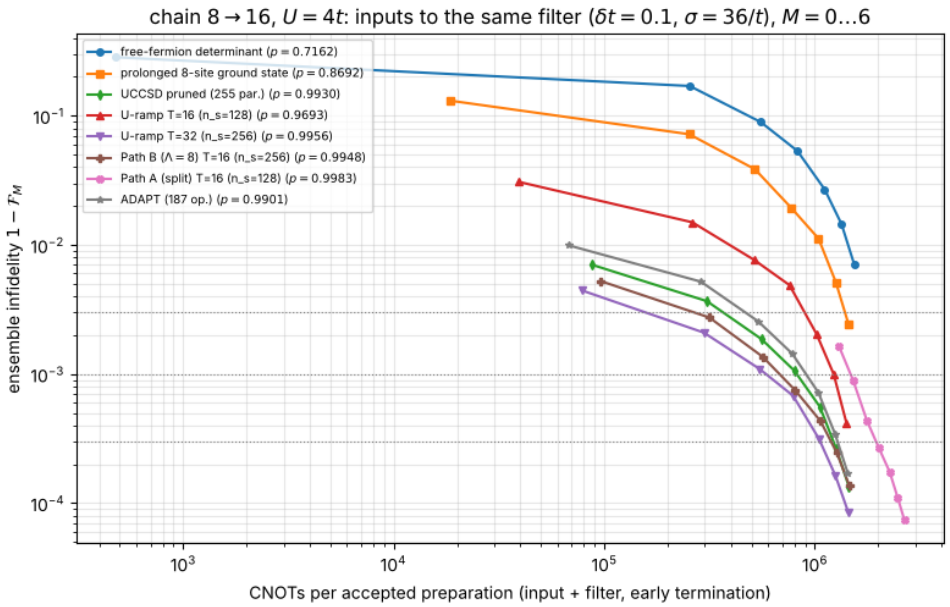


Figure 7: Cost versus accuracy for the chain $8 \to 16$ at $U = 4t$: ensemble infidelity against CNOTs per accepted preparation ($M = 0 \dots 6$) for the inputs of Table 3, all cleaned by the same Trotterized filter; dotted lines mark 3×10^{-3}, 10^{-3} and 3×10^{-4}.

σ and δt, a convergence test in T and n_s on the measured energy) and a comparison at matched shot budget including the training measurements of the variational routes are the missing pieces for a hardware run, for which the $4 \to 8$ chain and the $(1, 1)$ plaquette (thirty-two qubits, 256-dimensional sector) are within reach of current devices.

Acknowledgments

The work of C.G. and G.L. acknowledges funding from the European Union's Horizon Europe research and innovation program under the Marie Skłodowska-Curie Grant Agreement No. 101126636. M.H.-J. is partially supported by U.S. National Science Foundation (NSF) Grants PHY-1404159 and PHY-2310020. The authors acknowledge Anthropic's Claude for assistance with literature search and coding.

Code and data availability

All results are produced by the notebook `notebooks/ResolutionRefHubbard.ipynb`, built by `scripts/build_notebook_hubbard.py` on the module `pairinglib.hubbard` (lattices, sector ED, block prolongation, Galerkin coarse model, both paths, exact and Trotterized refinement, the ensemble rodeo formula and its Trotterized version, the symmetry diagnostics, the sector-level UCCSD and ADAPT-VQE, the Kronecker-structured product formulas of the sixteen-site study and the gate counting), which is covered by the test suite of the repository [18]; the figures are extracted from the executed notebook, and the Supplemental Material [20] and the notebook together document every number quoted. The ED reference code (Hubbard and t–J models; chain, square and checkerboard lattices) is the notebook `Hubbard.ipynb` [22], whose conventions the module follows.

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